



PLANT HORMONES — FULL MIND MAP

Biology Ch. 8.iii · Plant Physiology · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

1. Plant Hormones — Meaning, Examples & Cytokinin
2. Major Hormones & Functions — Auxin, Gibberellin, Cytokinin, ABA
3. Absciscic Acid (ABA) — Drought Tolerance & Stress Hormone
4. Ethylene — Gaseous Hormone & Natural Fruit Ripening
5. Artificial Fruit Ripening — Acetylene vs Ethylene
6. Parthenocarp — Seedless Tomato by Hormone Spray

1. Plant Hormones — Meaning, Examples & Cytokinin

- Plant hormones naturally occurring special organic compounds hote hain jo bahut kam quantity mein plant ke different parts tak transport hokar metabolic activities ko affect/control karte hain. Source examples: Auxin, Gibberellin, Cytokinin, Absciscic acid aur Ethylene. [Q2]
- Cytokinin ek plant hormone hai; source ke repeated MCQs mein Insulin, Thyroxine aur Estrogen ke opposite Cytokinin ko plant hormone identify kiya gaya hai. Cytokinin plant roots aur shoots mein cell division promote karta hai. [Q2, Q3, Q4]
- Insulin plant hormone nahi hai. Source ke mutabik ye pancreas ke islets of Langerhans se produce hota hai, blood glucose regulate karta hai aur insulin ki deficiency diabetes se linked hai. [Q1]

⚠ EXAM TRAP — Plant Hormone vs Animal Hormone

- ▶ Cytokinin = PLANT hormone; Insulin, Thyroxine, Estrogen = source questions mein plant hormones NAHI. [Q1–Q4]
- ▶ Plant-hormone examples source mein: Auxin, Gibberellin, Cytokinin, ABA, Ethylene. [Q2]

2. Major Hormones & Functions — Auxin, Gibberellin, Cytokinin, ABA

- Source matching ke according: Auxin → Apical dominance; Gibberellin → Cell elongation; Cytokinin → Cell division; Absciscic acid → Growth inhibition. [Q5]

⚠ EXAM TRAP — Hormone–Function Matching

- ▶ Auxin = Apical dominance | Gibberellin = Cell elongation. [Q5]
- ▶ Cytokinin = Cell division | Absciscic acid (ABA) = Growth inhibition. [Q5]

3. Absciscic Acid (ABA) — Drought Tolerance & Stress Hormone

- Absciscic acid drought tolerance se related hormone hai. Water shortage ke time ABA stomata ko temporarily close karwata hai, antitranspirant ki tarah act karta hai aur source ise “stress hormone” bhi kehta hai. [Q6]

4. Ethylene — Gaseous Hormone & Natural Fruit Ripening

- Ethylene gaseous form mein paaya jane wala phytohormone hai. Source ke mutabik ye fruit ripening, flower opening aur leaves ke abscission/shedding ko stimulate ya regulate karta hai. [Q7]
- Natural plant hormone jo mainly fruit ripening ke liye responsible hai, source ke according Ethylene hai. [Q10]

5. Artificial Fruit Ripening — Acetylene vs Ethylene

- Historical MCQ framing mein green fruits ke artificial ripening ke liye Acetylene answer diya gaya hai; calcium carbide acetylene ka source raha hai. Source explicitly note karta hai ki Acetylene option mein hone par historical MCQ context mein ise correct maana gaya. [Q8, Q9]

- Source explanation ye bhi distinguish karta hai ki Acetylene natural plant hormone nahi hai aur Ethylene jitna effective nahi; calcium carbide mein toxic arsenic/phosphorus traces ho sakte hain. Ethylene natural ripening hormone hai. [Q8, Q10]

⚠ EXAM TRAP — Acetylene vs Ethylene — Source-Specific

- ▶ Historical artificial-ripening MCQ answer = ACETYLENE (source/key framing). [Q8, Q9]
- ▶ Natural phytohormone mainly responsible for fruit ripening = ETHYLENE. [Q10]
- ▶ Source itself notes acetylene/calcium-carbide use restrictions and distinguishes it from natural ethylene. [Q8]

6. Parthenocarp — Seedless Tomato by Hormone Spray

- Seedless tomato fruits flowers par hormones spray karke produce kiye ja sakte hain. Source parthenocarp ko fertilization of ovules ke bina fruit production batata hai; Gibberellin, Auxin ya Cytokinin flower spray se artificial parthenocarp stimulate kar sakte hain. [Q11]